(Discrete) Distribution Flash Cards

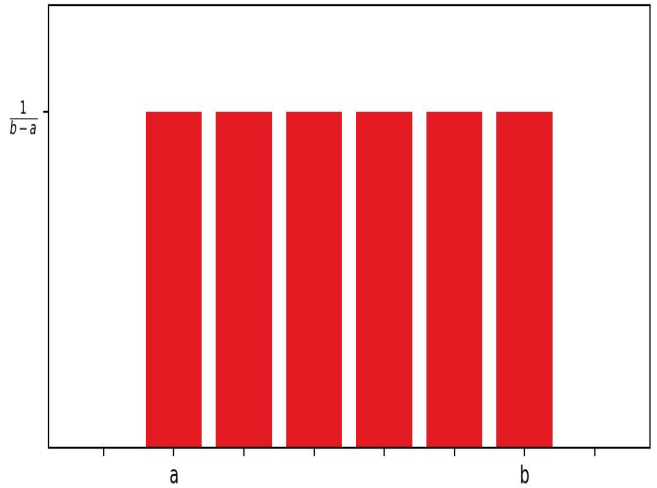
GENERAL ASSEMBLY

The Discrete Uniform Distribution

Used when all outcomes are equally likely.
Ex: Outcome of rolling a six-sided die might be U(1, 6)

pmf	$f(x) = \frac{1}{b-a}$
support	$a \leq x \leq b$
mean	$\frac{a+b}{2}$
variance	$\frac{(b-a)^2}{12}$

$$X \sim U(a, b)$$



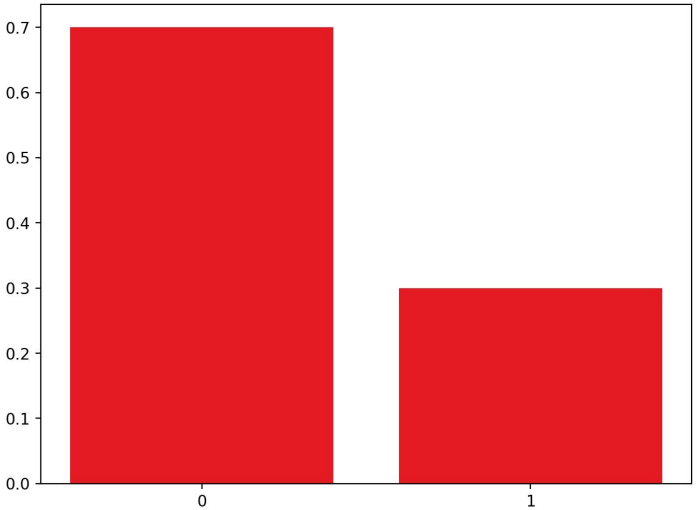
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The Bernoulli Distribution

Used when there are only two outcomes, and the “success” has fixed probability *p*
Ex: Outcome of flipping a fair coin is Ber(0.5)

pmf	$f(x) = p^x(1-p)^{1-x}$
support	$x \in \{0, 1\}$
mean	p
variance	$p(1-p)$

$$X \sim \text{Ber}(p)$$



This graph depicts Ber(0.3)

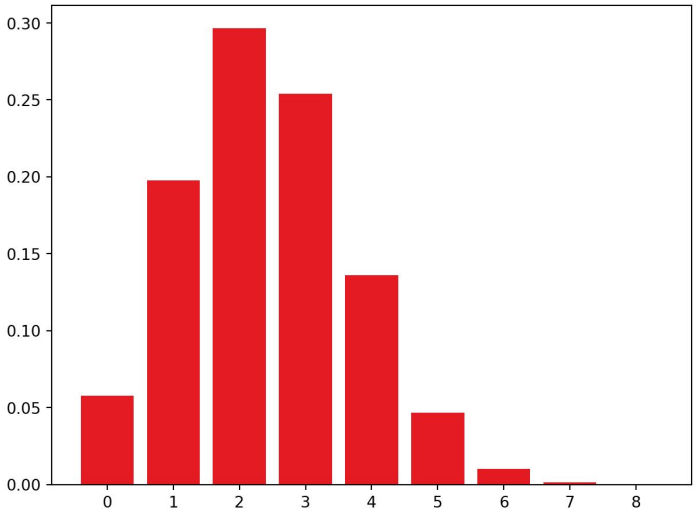
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The Binomial Distribution

Used when counting the number of successes when conducting *n* Bernoulli trials with fixed probability *p*.
Ex: Number of heads when flipping a fair coin 10 times is Bin(10, 0.5)

pmf	$f(x) = \binom{n}{k} p^x (1-p)^{n-x}$
support	$0 \leq x \leq n$
mean	np
variance	$np(1-p)$

$$X \sim \text{Bin}(n, p)$$



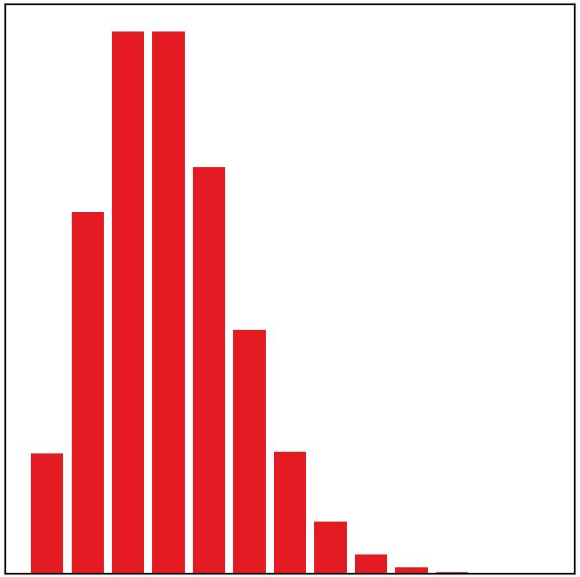
This graph depicts Bin(8, 0.3)

The Poisson Distribution

Used when estimating counts in a fixed interval (usually of time). *Ex*: If you generally receive 5 emails a day, you might model this distribution with Poi(5).

pmf	$f(x) = \frac{\lambda^x e^{-\lambda}}{x!}$
support	$x \geq 0$
mean	λ
variance	λ

$$X \sim \text{Poi}(\lambda)$$



Extras

Still valuable to know, but not as frequently used or relevant

The Geometric Distribution

Suppose you have a repeated Bernoulli trial event with probability of success *p*. The **geometric distribution** describes how many trials you have until your first success.

pdf	$f(x) = (1-p)^{x-1}p$
support	$x \geq 1$
mean	$\frac{1}{p}$
variance	$\frac{1-p}{p^2}$

Example: If ⅕ candies in a pack are green, the number of candies you eat until you randomly eat a green one might be Geometric(⅕).

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The Hypergeometric Distribution

Suppose you take samples from a fixed population where

- N = population size
- K = successes in population
- n = sample size

Then the number of successes you see in your sample might follow Hypergeom(N, K, n).

Example: In a game of poker, you draw 5 cards. The number of aces in your hand would follow Hypergeom(N = 52, K = 4, n = 5).

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The Multinomial Distribution

Consider a special Bernoulli trial in which there were more than two possible outcomes. Because we're not dealing with just 0s and 1s anymore, we have our first **random vector**!

Example: Suppose it's known the American population's voting registry is 40% Democrat, 40% Republican, and 20% Independent/Other. If we go out and sample 100 people, the distribution of our results might be:

$$\mathbf{X} \sim \text{Multinomial}\left(100, \begin{bmatrix} 0.4 \\ 0.4 \\ 0.2 \end{bmatrix}\right)$$

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